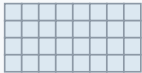


Input CSI

$$H \in \mathbb{C}^{T \times K \times N_h \times N_v}$$



N_h : rows
 N_v : columns

Per-Antenna TF Embedding



Shared TF
Embedding E_θ

Shared across antennas
No early antenna mixing

Geometry- Preserving Tokens



$$z_{t,k,r,c}$$

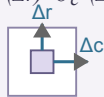
$$(t, k, r, c)$$

Explicit UPA coordinates

UPA Geometry-Aware Transformer

Self-Attention
+ UPA Relative Bias

$$B_h(\Delta r, \Delta c) = b_r^h(\Delta r) + b_c^h(\Delta c)$$



Masked Reconstruction & Prediction

Random · Temporal
Frequency



Antenna Row Column 2D
Block

$\hat{H}$ Reconstruction · Prediction
Partial-Array Recovery

Geometry-Preserving Representation

2D Relative Geometry

UPA-Structured Pretraining